

Basketball Analytics: Core Formula Cheat Sheet

A concise collection of widely used basketball metrics with brief explanations. Formulas are expressed in plain text for clarity.

Shooting Efficiency

Field Goal Percentage (FG%): $FG\% = FGM / FGA$

Three-Point Percentage (3P%): $3P\% = 3PM / 3PA$

Free Throw Percentage (FT%): $FT\% = FTM / FTA$

Effective Field Goal Percentage (eFG%): $eFG\% = (FGM + 0.5 \times 3PM) / FGA$ — credits the added value of made threes.

True Shooting Percentage (TS%): $TS\% = PTS / (2 \times (FGA + 0.44 \times FTA))$ — overall scoring efficiency including free throws.

Free Throw Rate (FTr): $FTr = FTA / FGA$ — how often a team/player gets to the line relative to shot attempts.

3-Point Attempt Rate (3PA Rate): $3PA \text{ Rate} = 3PA / FGA$ — shot diet skew toward threes.

Points per Shot (PPS): $PPS = PTS \text{ from FGs} / FGA$ — average points per field-goal attempt.

Possessions & Pace

Team Possessions (estimate): $Poss = 0.5 \times [(FGA + 0.44 \times FTA - ORB + TOV) + (Opp \text{ FGA} + 0.44 \times Opp \text{ FTA} - Opp \text{ ORB} + Opp \text{ TOV})]$ — widely used estimator.

Offensive Possessions (team, alt): $OffPoss \approx FGA + 0.44 \times FTA - ORB + TOV$

Pace (per game): $Pace = 48 \times (Team \text{ Poss} + Opp \text{ Poss}) / (2 \times \text{Minutes per team})$ — possessions per 48 minutes.

Turnovers

Team Turnover Rate (TOV%): $TOV\% = TOV / Poss$ — share of possessions ending in turnovers.

Player Turnover Rate (player TOV%): $\text{player TOV}\% = TOV / (FGA + 0.44 \times FTA + TOV)$ — how often individual plays end in a turnover.

Rebounding

Offensive Rebound Percentage (ORB%): $ORB\% = ORB / (ORB + Opp \text{ DRB})$ — share of available offensive rebounds secured.

Defensive Rebound Percentage (DRB%): $DRB\% = DRB / (DRB + Opp \text{ ORB})$ — share of available defensive rebounds secured.

Total Rebound Percentage (TRB%): $TRB\% = TRB / (TRB + Opp \text{ TRB})$ — share of all available rebounds.

Ratings (Per 100 Possessions)

Offensive Rating (OffRtg): $OffRtg = 100 \times \text{Points Scored} / Possessions$ — points scored per 100 possessions.

Defensive Rating (DefRtg): $DefRtg = 100 \times \text{Opponent Points} / \text{Opponent Possessions}$ — points allowed per 100 possessions.

Net Rating (NetRtg): $NetRtg = OffRtg - DefRtg$ — point differential per 100 possessions.

Shot Profile & Expected Value

Expected Points (single shot type): $xPts = Prob(\text{make}) \times PointsIfMade$ — expected value of a shot type/zone.

Shot Mix Expected Value: $xPts_total = \sum_i [share_i \times (make_i \times points_i)]$ — aggregate expected efficiency from shot distribution.

Effective Shot Quality (example): $ESQ = \sum_z [attempts_z \times (make_prob_z \times points_z)] / \sum_z attempts_z$ — efficiency implied by shot locations/types.

Four Factors (Team Level)

Four pillars influencing efficiency: Shooting (eFG%), Turnovers (TOV%), Rebounding (ORB%), Free Throws (FTr).

Typical emphasis: shooting ~40%, turnovers ~25%, rebounding ~20%, free throws ~15% — indicative weights used in analysis.

Lineups & On/Off Concepts

On/Off Net Impact: $\text{On/Off} = \text{NetRtg (on court)} - \text{NetRtg (off court)}$ — context dependent; not a pure causal estimate.

Lineup Net Rating: $\text{NetRtg_lineup} = 100 \times (\text{Points_for_lineup} - \text{Points_against_lineup}) /$

Poss_lineup — strength of a specific group.

Pythagorean Expectation

$\text{Win\%} \approx (\text{Points For}^x) / (\text{Points For}^x + \text{Points Against}^x)$ — exponent x is sport/era specific; NBA often ~13–14.

Expected Wins: $\text{ExpWins} = \text{Win\%} \times \text{Games}$.

Free Throws & And-1s

Free Throws per Field-Goal Attempt: FTA/FGA — whistle frequency proxy.

And-1 Rate: $\text{And1\%} = \text{And1Makes} / \text{FGMs}$ — frequency of made shots with a foul leading to one FT.

Miscellaneous

Points per Possession (PPP): $\text{PPP} = \text{Points} / \text{Possessions}$ — use for plays, sets, or players (when possessions are attributed).

Offensive Rebound Putback Rate: $\text{Putback\%} = \text{Team Putback FGAs} / \text{Team ORB}$ — how often ORBs become immediate shots.

Transition Rate: $\text{Trans\%} = \text{Transition Plays} / \text{Total Plays}$ — share of possessions in transition.

Foul Draw Rate: $\text{Foul Draw\%} = \text{Shooting Fouls Drawn} / \text{FGA}$ — how often a shooter draws a foul per attempt.

Notes & Usage

These formulas are commonly used across the basketball analytics community. Exact implementations can vary by data source (e.g., possession attribution, play-by-play parsing, or video tag definitions). Always document your definitions, confirm with your dataset provider, and check calibration (e.g., sum of lineup possessions vs. team totals).